

Fig. 2: An actual-size, single-side PCB for the simple square-wave generator

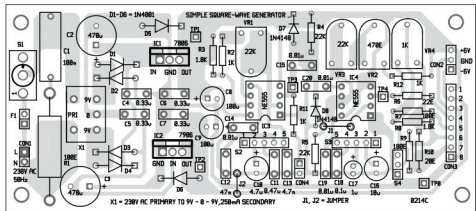


Fig. 3: Component layout for the PCB

depending on the position of switch S3. Duration of the positive pulse produced by timer IC4 depends on the position of switch S3 as indicated in Table II.

The pulse ranges produced by IC4 can be changed according to the field of application of the generator. The duration of the pulses is adjusted roughly with the potentiometer VR3 and more precisely with VR2. The output level of the IC4 can be controlled with potentiometer VR4.

The output of IC3 provides different frequencies at CON4 depending on the position of switch S2. The output of IC4 can drive loads in many configurations. Several possible combinations with pins of connector CON3 may be used, such as:

1. Load can be connected between main output pin 4 and pin 1 (+5V) of CON3.
2. Load can be connected between main output pin 4 and pin 2 (-5V) of CON3.

3. Load can be connected between main output pin 4 and ground pin 3 of CON3.

4. Two loads can be connected between main output pin 4 and positive and negative power supply in pin 1 and pin2 of CON3.

The minimal load at pin 4 of CON3 in all configurations is preferably above 100-ohm. The minimal load at pins 5, 6 and 7 of CON3 in all configurations is preferably above 2-kilo-ohm (depends on the required accuracy for the adjustment of the output amplitude). Short circuits between the outputs and the ground will not damage the circuit but should be avoided. You can add protecting diodes 1N400X between the output pin 3 of the IC4 and +6V and -6V if you drive reactive loads.

Switch S4 switches the junction of output of resistive divider and one end of potentiometer VR4 to ground or to the negative power supply.

In brief, using this circuit, you can

PARTS LIST

Semiconductors:

- IC1 - 7806, +6V regulator
- IC2 - 7906, -6V regulator
- IC3, IC4 - NE555 timer
- D1-D6 - 1N4001 rectifier diode
- D7-D8 - 1N4148 signal diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise):

- R1 - 100-ohm, 2W
- R2, R11, R12 - 1-kilo-ohm
- R3, R8 - 1.8-kilo-ohm
- R4, R5 - 22-kilo-ohm
- R6 - 22-ohm
- R7 - 100-ohm
- R9 - 180-ohm
- R10 - 20-ohm
- VR1, VR3 - 22-kilo-ohm potmeter
- VR2 - 470-ohm potmeter
- VR4 - 1-kilo-ohm potmeter

Capacitors:

- C1 - 100nF, 630V polyester
- C2, C3 - 470uF, 35V electrolytic
- C4-C7 - 0.33uF, ceramic disk
- C8, C9 - 100uF, 16V electrolytic
- C10 - 4.7uF, 16V electrolytic
- C11 - 0.47uF, ceramic disk
- C12 - 47nF, ceramic disk
- C13 - 4.7nF, ceramic disk
- C14, C15, C19, C20 - 0.01uF, ceramic disk
- C17 - 1uF, 16V electrolytic
- C16 - 10uF, 16V electrolytic
- C18 - 0.1uF, ceramic disk

Miscellaneous:

- F1 - 250mA fuse
- X1 - 230V AC primary to 9V-0-9V, 250mA secondary transformer
- CON1, CON4 - 2-pin connector
- CON2 - 3-pin connector
- CON3 - 8-pin connector
- S1 - On/off switch
- S2, S3 - 4-way rotary switch
- S4 - 2-way rotary switch

take +6V, 0, -6V at CON2, measure and use different frequencies at CON4. CON3 is used to connect different loads as mentioned above.

Construction and testing

An actual-size, single-side PCB for the simple square-wave generator is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on PCB, enclose it in a suitable plastic case. Fix a 3-pin connector for +6V, -6V and ground and fix another 8-pin connector for different wave outputs at the front side of the case. Also fix a 2-pin connector for mains power supply on the back side of the panel.

To check the circuit for proper functioning, verify 6V supply at TP1 and -6V at TP2, with respect to TP0. ●

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